PYTHON FOR ACTUARIES · NOTEBOOK 09 · EXERCISE

Motor Insurance Fraud Detection

Final Exercise

Build a classifier that detects fraudulent motor insurance claims. Work through three phases:

- **Phase 1**: Exploration
- **Phase 2**: Preparation and Modelling
- **Phase 3**: Evaluation

Phases 1 and 2 repeat steps from the exercises in Notebooks 05 and 07 on purpose: here they come together in one model. Your code from there can serve as a starting point.

The Dataset: `vehicle_fraud.csv`

10,050 rows of synthetic motor insurance claims:

Column	Description
`claim_id`	Claim number
`policyholder_age`	Age in years
`occupation`	`Employed`, `Self-employed`, `Unemployed`, `Student`
`region`	Region `R01` to `R10`
`credit_score`	Credit score (300–850)
`customer_since_years`	Years as a customer
`vehicle_value`	Vehicle value
`vehicle_type`	`Compact`, `Mid-range`, `Luxury`, `SUV`, `Sports car`
`coverage_type`	`Liability`, `Partial coverage`, `Comprehensive`
`number_of_prior_claims`	Number of earlier claims
`claim_amount`	Claim amount
`police_report`	`1` if a police report was filed
`witnesses_present`	`1` if witnesses were present
`reporting_delay_days`	Days between accident and claim report
`fraud`	Target: `1` = fraudulent, `0` = legitimate

```
# Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier
from xgboost import XGBClassifier
from sklearn.metrics import (accuracy_score, precision_score, recall_score, f1_score,
                             roc_auc_score, confusion_matrix, precision_recall_curve)
from sklearn.inspection import permutation_importance

# Seaborn style for all plots
sns.set_theme(style='whitegrid')

# Avoid truncation of DataFrame output in Jupyter
pd.set_option('display.expand_frame_repr', False)

# Display all columns in the DataFrame output
pd.set_option('display.max_columns', None)

# Define a separator for better readability
sep = "=" * 50
```

```
# Load the dataset
df = pd.read_csv('vehicle_fraud.csv')
df.head()
```

	claim_id	policyholder_age	occupation	region	credit_score	customer_since_years	vehicle_value	vehicle_type	coverage_type	number_of_prior_claims	claim_amount	police_report	witnesses_present	reporting_delay_days	fraud
0	108777	38	Unemployed	R05	575.0	14	75699.57	Luxury	Liability	2	14926.72	0	0	31	0
1	102200	42	Employed	R10	653.0	21	70254.54	Sports car	Partial coverage	2	4112.72	1	0	16	0
2	109597	27	Self-employed	R09	525.0	2	9686.20	Compact	Comprehensive	2	5651.48	0	0	44	0
3	108491	43	Employed	R03	619.0	14	77918.56	Compact	Comprehensive	2	6838.21	0	1	31	0
4	103315	45	Student	R04	624.0	8	3660.94	Luxury	Partial coverage	4	30278.60	0	0	33	1

Phase 1: Exploration

Get to know the data: size, missing values, fraud rate, distributions and the first fraud patterns.

This exploration is deliberately short and focuses on fraud. The general exploration of the data took place in Notebook 06.

1.1 Overview

Task 1: How many rows and columns does the dataset have? Which columns are numeric, which are text?

```
# Task 1: Shape and data types
# Hint: df.shape, df.info()
```

Task 2: Are there missing values?

```
# Task 2: Missing values
# Hint: df.isna().sum()
```

Task 3: What is the fraud rate?

```
# Task 3: Fraud rate
# Hint: df['fraud'].value_counts(normalize=True)
```

1.2 Distributions

Task 1: Plot a histogram of `claim_amount`. Is it symmetric or skewed?

```
# Task 1: Histogram of claim_amount
# Hint: df['claim_amount'].plot(kind='hist', bins=50)
#       plt.title('...'), plt.xlabel('...'), plt.ylabel('...')
#       plt.show()
```

Task 2: Compare `claim_amount` for fraudulent and legitimate claims with a box plot.

```
# Task 2: Box plot of claim_amount by fraud
# Hint: sns.boxplot(data=df, x='fraud', y='claim_amount')
```

1.3 Fraud Patterns

Task 1: Compare the mean of every numeric column for `fraud == 0` and `fraud == 1`.

```
# Task 1: Means by fraud status
# Hint: df.groupby('fraud').mean(numeric_only=True)
```

Task 2: Compute the correlation of every numeric column with `fraud` and sort it.

```
# Task 2: Correlation with fraud
# Hint: df.corr(numeric_only=True)['fraud'].sort_values()
```

Task 3: Compute the fraud rate for each combination of `police_report` and `witnesses_present`.

```
# Task 3: Fraud rate by documentation
# Hint: df.groupby(['police_report', 'witnesses_present'])['fraud'].mean()
```

Task 4: Compute the fraud rate for new customers (`customer_since_years < 3`) with many prior claims (`number_of_prior_claims >= 3`).

```
# Task 4: Fraud rate of new customers with many prior claims
# Hint: new_many = (df['customer_since_years'] < 3) & (df['number_of_prior_claims'] >= 3)
#       df.groupby(new_many)['fraud'].mean()
```

Phase 2: Preparation and Modelling

Apply Notebooks 05 and 07: clean, engineer features, encode, split, fill, scale and train one or more models.

2.1 Cleaning

Task 1: How many rows are duplicates? Remove them.

```
# Task 1: Duplicates
# Hint: df.duplicated().sum()
#       df = df.drop_duplicates()
```

Task 2: Find the impossible ages with `describe()` and replace them with `np.nan`.

```
# Task 2: Impossible ages
# Hint: df['policyholder_age'].describe()
#       df.loc[df['policyholder_age'] < 18, 'policyholder_age'] = np.nan
```

Task 3: Add the indicator column `credit_score_missing`. Compare the fraud rate of claims with and without `credit_score`.

```
# Task 3: Indicator column and fraud rate
# Hint: df['credit_score_missing'] = df['credit_score'].isna().astype(int)
#       df.groupby('credit_score_missing')['fraud'].mean()
```

2.2 Feature Engineering

Create four new columns:

Task 1: `claim_value_ratio` = `claim_amount` / `vehicle_value`

```
# Task 1: Claim-to-value ratio
# Hint: df['claim_value_ratio'] = df['claim_amount'] / df['vehicle_value']
```

Task 2: `no_report_no_witness` = `1` if there is neither a police report nor witnesses

```
# Task 2: Neither police report nor witnesses
# Hint: ((df['police_report'] == 0) & (df['witnesses_present'] == 0)).astype(int)
```

Task 3: `new_customer_many_claims` = `1` for customers with `customer_since_years < 3` and `number_of_prior_claims >= 3`

```
# Task 3: New customer with many prior claims
```

Task 4: `log_claim_amount` = `np.log(claim_amount)` for the skewed claim amounts

```
# Task 4: Log-transformed claim amount
```

Then compare the mean `claim_value_ratio` for fraudulent and legitimate claims.

```
# Mean claim_value_ratio by fraud status
# Hint: df.groupby('fraud')['claim_value_ratio'].mean()
```

2.3 Encoding, Split, Missing Values and Scaling

Task 1: One-hot encode `occupation`, `region`, `vehicle_type` and `coverage_type` with `pd.get_dummies()`. Store the result in `df_encoded`.

```
# Task 1: One-hot encoding
# Hint: pd.get_dummies(df, columns=[...], drop_first=True, dtype=int)
```

Task 2: Split into features `X` (without `claim_id` and `fraud`) and target `y`.

```
# Task 2: Features and target
# Hint: X = df_encoded.drop(columns=['claim_id', 'fraud'])
#       y = df_encoded['fraud']
```

Task 3: Split into 80% training and 20% test data. `stratify=y` keeps the fraud rate equal in both sets.

```
# Task 3: Train/test split
# Hint: train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Task 4: Fill the missing values in both sets with the medians of the training data.

```
# Task 4: Fill missing values with the training medians
# Hint: medians = X_train.median()
#       X_train = X_train.fillna(medians)
#       X_test = X_test.fillna(medians)
```

Task 5: Scale the features with `StandardScaler`. Fit the scaler on the training data only.

```
# Task 5: Scaling
# Hint: scaler = StandardScaler()
#       X_train_scaled = scaler.fit_transform(X_train)
#       X_test_scaled = scaler.transform(X_test)
```

2.4 Training

Train one or more of the following classifiers.

With only 12% fraud, a model mostly learns the legitimate claims. Class weighting counteracts this: each fraud case counts as much as about 7.5 legitimate claims (number of legitimate / number of fraudulent claims), so both classes carry the same total weight. `class_weight='balanced'` and `scale_pos_weight` do the same thing; `XGBClassifier` just calls it differently. `MLPClassifier` has no class weighting.

Model	Input	Class weighting
`LogisticRegression(max_iter=1000)`	scaled	`class_weight='balanced'`
`RandomForestClassifier(n_estimators=100)`	unscaled	`class_weight='balanced'`

Model	Input	Class weighting
`XGBClassifier(n_estimators=100)`	unscaled	`scale_pos_weight` = number of legitimate / number of fraudulent claims
`MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, early_stopping=True)`	scaled	none

Use `random_state=42` for every model.

```
# Train one or more models
# Hint: lr_model = LogisticRegression(max_iter=1000, class_weight='balanced',
    random_state=42)
#       lr_model.fit(X_train_scaled, y_train)
# Hint: for XGBoost, scale_pos_weight = (y_train == 0).sum() / (y_train == 1).sum()
```

Phase 3: Evaluation

Compare the models with metrics, find the important features and pick the cheapest threshold.

3.1 Metrics

Metric	Question
Precision	Of the flagged claims, how many are fraud?
Recall	Of all fraud cases, how many are flagged?
F1-score	Balance of precision and recall
ROC-AUC	How well does the model rank fraud above non-fraud?

Store your trained models with their test data in the dictionary `models`. Keep only the entries of the models you trained.

```
# Uncomment the entries of the models you trained
models = {
    # 'Logistic Regression': (lr_model, X_test_scaled),
    # 'Random Forest': (rf_model, X_test),
    # 'XGBoost': (xgb_model, X_test),
    # 'Neural Network': (mlp_model, X_test_scaled),
}
```

Task 1: For each model, compute accuracy, precision, recall, F1-score and ROC-AUC. Show the results as a DataFrame.

```
# Task 1: Metrics table
# Hint: results = {}
#       for name, (model, X_eval) in models.items():
#           y_pred = model.predict(X_eval)
#           y_proba = model.predict_proba(X_eval)[:, 1]
#           results[name] = {'Accuracy': accuracy_score(y_test, y_pred), ...,
#                            'ROC-AUC': roc_auc_score(y_test, y_proba)}
#       results_df = pd.DataFrame(results).T
```

Task 2: Plot a confusion matrix for each model.

```
# Task 2: Confusion matrices
# Hint: for name, (model, X_eval) in models.items():
#           cm = confusion_matrix(y_test, model.predict(X_eval))
#           sns.heatmap(data=cm, annot=True, fmt='d', cmap='Blues')
#           plt.title(name), plt.xlabel('Predicted'), plt.ylabel('Actual')
#           plt.show()
#       fmt='d' shows whole numbers
```

3.2 Feature Importance

`permutation_importance()` shuffles one feature at a time in the test data and measures how much the ROC-AUC drops. A large drop means the model relies on that feature.

Task 1: Pick one model from `models` and compute its permutation importance.

```
# Task 1: Permutation importance
# Hint: name = 'Random Forest'
#       model, X_eval = models[name]
#       perm = permutation_importance(model, X_eval, y_test, n_repeats=10,
#                                     scoring='roc_auc', random_state=42)
```

Task 2: Plot the ten most important features as a horizontal bar chart.

```
# Task 2: Top 10 features
# Hint: importances = pd.Series(perm.importances_mean, index=X_test.columns)
#       top10 = importances.sort_values().tail(10)
#       top10.plot(kind='barh')
```

Task 3: Do your new columns from 2.1 and 2.2 appear among them?

3.3 Threshold Tuning

`predict()` flags a claim as fraud if its predicted probability is at least 0.5. A lower threshold catches more fraud (higher recall) but causes more false alarms (lower precision). The best threshold depends on the costs:

- Every flagged claim is investigated. An investigation costs 500 €.
- Every missed fraud case is paid out. It costs its `claim_amount`.

$$\text{Cost}(t) = 500 \cdot \text{number of flagged claims} + \text{sum of claim amounts of missed fraud cases}$$

Task 1: For the model from 3.2, compute precision and recall for every threshold with `precision_recall_curve()` and plot recall against precision.

```
# Task 1: Precision-recall curve
# Hint: y_proba = model.predict_proba(X_eval)[:, 1]
#       precisions, recalls, thresholds = precision_recall_curve(y_test, y_proba)
#       pr_curve = pd.Series(precisions, index=recalls)
#       pr_curve.plot(kind='line')
```

Task 2: Compute the total cost for the thresholds 0.01, 0.02, ..., 0.99 and plot cost against threshold. Which threshold has the lowest cost? Compare it with 0.5.

```
# Task 2: Total cost per threshold
# Hint: def cost_table(investigation_cost):
#           rows = []
#           for t in np.arange(1, 100) / 100:
#               flagged = y_proba >= t
#               missed = ~flagged & (y_test == 1)
#               cost = investigation_cost * flagged.sum() + X_test.loc[missed,
    'claim_amount'].sum()
#               rows.append({'threshold': t, 'flagged': flagged.sum(),
#                            'recall': recall_score(y_test, flagged), 'cost': cost})
#           return pd.DataFrame(rows)
# Hint: cost_curve = costs.set_index('threshold')['cost']
#       cost_curve.plot(kind='line')
#       costs.loc[costs['cost'].idxmin()]
```

Task 3: Repeat Task 2 with investigation costs of 200 € and 3000 €. How does the best threshold move?

```
# Task 3: Investigation costs of 200 € and 3000 €
```

Summary

- **Exploration:** fraud rate, skewed distributions, fraud rates by group
- **Preparation and modelling:** duplicates, impossible values, indicator for missing values, feature engineering, one-hot encoding, stratified split, filling missing values, scaling, class weights
- **Evaluation:** precision, recall, F1-score, ROC-AUC, confusion matrix, permutation importance, cost-based threshold tuning
- scikit-learn User Guide
- XGBoost Documentation